

RIDDER, Kazakhstan

WMO: 362080

Lat: 50.333N

Lon: 83.550E

Elev: 2661

StdP: 13.34

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.6	-19.5	-31.0	1.1	-26.3	-24.1	1.6	-19.0	37.3	31.4	32.2	31.2	2.6	90	0.586

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.9	83.7	63.9	80.6	62.7	77.8	61.6	66.5	78.9	64.6	77.0	62.9	75.0	5.0	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.1	92.2	72.8	59.9	84.9	70.8	58.0	79.3	69.1	32.8	79.0	31.2	76.8	29.9	75.1	75.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.6	19.3	14.2	-31.5	87.9	8.1	3.1	-37.4	90.2	-42.1	92.0	-46.7	93.8	-52.6	96.1		
(5)			-31.7	70.5	8.1	2.8	-37.5	72.6	-42.2	74.2	-46.8	75.8	-52.6	77.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.3	8.7	13.0	24.9	39.8	51.1	60.5	63.0	60.4	50.3	39.0	23.3	12.5		
	DBStd	21.75	13.19	12.45	11.66	9.78	7.99	6.56	5.38	6.14	8.26	9.46	12.86	12.32		
	HDD50	5934	1281	1037	779	321	87	7	0	3	97	355	803	1164		
	HDD65	10197	1746	1457	1244	757	434	159	101	168	445	805	1252	1629		
	CDD50	1306	0	0	0	15	120	322	404	325	104	15	1	0		
	CDD65	92	0	0	0	0	2	24	39	24	3	0	0	0		
	CDH74	1592	0	0	0	2	85	409	514	476	104	2	0	0		
	CDH80	319	0	0	0	0	16	62	111	108	22	0	0	0		
	WSAvg	4.5	4.5	4.6	4.8	4.9	5.3	3.6	3.4	3.5	4.2	5.0	5.0	4.8		
	PrecAvg	25.2	0.7	0.8	1.3	2.3	3.0	3.0	3.5	2.6	2.1	2.4	2.1	1.4		
(15) Precipitation	PrecMax	32.4	1.4	1.5	2.5	5.6	7.5	6.0	6.3	5.7	5.9	4.7	5.4	3.0		
	PrecMin	15.6	0.1	0.3	0.4	0.4	0.8	0.8	1.5	0.0	0.4	0.8	0.9	0.2		
	PrecStd	4.4	0.4	0.4	0.5	1.3	1.6	1.3	1.3	1.3	1.3	1.0	1.0	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.7	43.3	56.5	72.5	82.0	84.9	87.7	87.8	82.9	70.9	57.4	43.5		
		MCWB	33.9	34.4	42.7	52.1	59.3	64.7	67.7	65.1	61.0	53.0	45.2	34.2		
	2%	DB	34.9	38.4	51.3	66.7	75.6	81.6	83.2	83.3	76.6	65.5	50.3	37.4		
		MCWB	28.6	31.2	39.7	49.4	56.9	62.9	65.0	63.1	57.6	50.4	40.4	31.1		
	5%	DB	29.8	34.3	46.5	62.1	71.7	78.8	79.7	79.3	72.0	59.9	44.7	32.2		
		MCWB	25.5	28.7	37.3	47.5	54.8	61.4	64.1	61.8	55.6	47.3	37.9	27.8		
	10%	DB	25.7	30.3	41.9	57.5	67.6	75.5	76.5	75.9	67.2	54.2	40.0	28.5		
		MCWB	22.3	26.0	34.4	44.9	52.7	60.4	63.0	60.7	53.4	44.5	34.7	25.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.7	35.3	43.5	53.8	62.9	68.1	70.4	68.3	62.1	54.7	46.2	35.2		
		MCDB	40.0	41.6	53.6	68.4	77.1	80.2	81.9	80.4	79.9	67.7	55.2	42.6		
	2%	WB	29.2	31.9	40.6	50.8	58.5	65.2	67.4	64.8	58.8	50.9	41.3	31.6		
		MCDB	34.3	37.6	49.4	65.1	72.9	77.9	79.3	79.3	73.1	63.6	48.9	36.7		
	5%	WB	25.7	29.0	37.8	48.3	56.3	63.2	65.3	63.0	56.4	48.2	38.2	28.2		
		MCDB	29.6	33.8	45.7	60.3	68.7	75.5	76.7	76.9	70.2	58.3	44.3	31.7		
	10%	WB	22.4	26.3	35.0	45.7	53.9	61.4	63.5	61.3	54.0	45.3	34.7	25.2		
		MCDB	25.6	30.4	41.2	56.0	65.3	73.1	74.3	73.9	66.1	53.7	39.9	28.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.3	21.5	21.7	21.4	23.0	22.7	20.9	23.9	24.1	19.3	17.1	17.1		
		MCDBR	19.6	21.9	25.0	30.6	30.6	28.6	27.4	30.8	33.3	30.4	21.6	19.1		
		MCWBR	15.0	16.1	16.1	17.3	16.4	14.4	14.0	15.4	17.4	17.4	13.9	14.3		
		MCDBR	18.9	20.5	22.7	28.5	27.2	25.3	24.2	27.5	30.8	28.4	20.9	18.3		
	5% WB	MCWBR	14.7	15.3	15.3	17.1	16.0	14.7	14.0	15.3	17.2	17.0	13.8	14.0		
(40) Clear-Sky Solar Irradiance	taub		0.279	0.311	0.362	0.418	0.403	0.401	0.423	0.388	0.350	0.345	0.310	0.282		
	taud		2.067	2.012	1.927	1.984	2.173	2.255	2.205	2.285	2.347	2.218	2.081	2.014		
	Ebn at Noon		230	252	259	259	271	272	263	267	266	240	213	204		
	Edh at Noon		26	36	48	51	44	41	43	38	32	30	26	24		
(44) All-Sky Solar Radiation	RadAvg		460	810	1230	1463	1839	1993	1844	1709	1246	672	388	328		
	RadStd		32	38	113	136	152	173	147	116	117	93	41	40		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)		N/A	N/A	N/A	N/A	N/A	-1.06	N/A	N/A	N/A	N/A

Nomenclature: See separate page